

THE CHEMICAL ENGINEERING CURRICULUM—1994

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The most recent survey of the chemical engineering undergraduate curricula, conducted every five years since 1957, was made by the Undergraduate Curriculum Subcommittee of the Education Projects Committee of the American Institute of Chemical Engineers (AIChE) in the summer of 1994.^[1] A questionnaire that closely corresponds with ABET/AIChE categories was sent to the 158 chemical engineering departments listed with AIChE; sixty-three departments responded, and the survey results are based on those responses.

The spreadsheets that contained the data collected were prepared using Quattro Pro (Borland Int'l.) to assist in analyzing the survey results. Table 1 summarizes the responses received. As shown, seventeen Canadian schools were sent the questionnaire, but only one responded. This school's data was difficult to translate into reasonably related US numbers, so it was not included in the report. The data available from US schools was reported at the Department Heads' Meeting of the AIChE annual meeting in Miami Beach, Florida, in November of 1995.



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TABLE 1
Summary of Responses Received

	United States	Canada
Total Schools Surveyed	158	17
Total Responses Received	63	1
Percent Responses	39.9%	5.9%
Percent Responses Overall	36.6%	

SURVEY RESULTS AND DISCUSSION

The semester credit hours required for the Bachelor's Degree remains almost the same as it was in the previous 1989 survey (when ninety-two schools responded). Figure 1 shows that the trend seems to have stabilized in the low 130s. The detailed information on the spreadsheet, however, indicates that semester hours actually range from 115 to 145. The lower bound has increased only slightly since the last survey. More than 80% of the departments require 125 to 140 semester hours, with only six having fewer than 125 and five

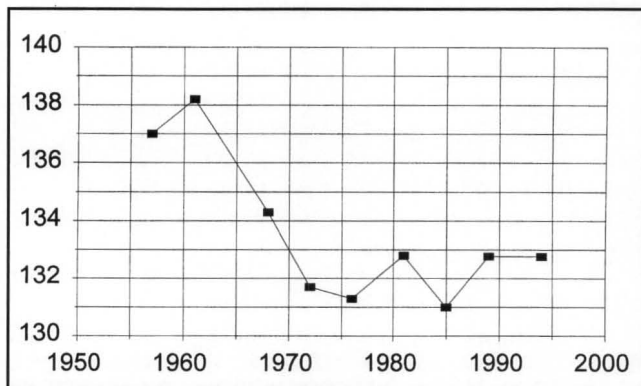


Figure 1. Total semester hours required for undergraduate chemical engineering degree.

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having more than 140. Most schools operate on the *Nominal Semester* basis; few use a quarter system.

As in the previous survey, the average curricular area distribution continue to be somewhat close to ABET/AIChE requirements. A closer look at the individual departmental requirements reveals a wide range (see Table 2). For example, mathematics, which has an average value of 16.5 contact hours (semester credits), ranges from 12 to 22 hours. Most of the departments' requirements fall within 15 to 18 hours. Expressed as a percentage, the mathematics requirement is 12.5%, equaling the AIChE requirement of 12.5%.

Similar traits are observed in other categories. For ex-

ample, "engineering science/design" and "humanities/social sciences" continue to increase their share and move away from ABET/AIChE requirements at the expense of other categories that show a downward movement.

Some changes can be observed within the categories. Mathematics, which used to be predominantly calculus and differential equations, show that calculus has maintained its dominance, but differential equations has lost some of its share to analytical geometry and advanced calculus.

The remaining credits still demonstrate wide diversity. The popularity of linear algebra has increased to 26 departments (41%) requiring the course, compared to 20 departments (22%) in 1989. Partial differential equations show reduced popularity. Many departments require a mathematics elective.

In the fundamentally important "basic science" category, introductory physics and chemistry continue to maintain diversity and dominate the credit hours. But ten departments reported modern physics, five listed biology, and four indicated that other basic sciences are required. A comparison of these numbers to previous survey results indicates that the popularity of these courses is decreasing. The advanced chemistry requirements showed a drop to 11.95% from the AIChE requirement of 12.5%, but, as usual, showed a wide range of 10.67 to 21.0 hours. The total chemistry contents, as indicated by Figure 2, has maintained a stable trend.

The engineering science and design requirements increased to 39.9% from their 39.7% value in the 1989 survey. The statics course has suffered a significant drop in popularity, with nearly 68% of the departments offering the course in 1989 and only about 59% in 1994. But dynamics and/or mechanics of materials maintain their positions, with approximately one-quarter of the departments requiring the courses.

Another startling change is seen in the introduction to electrical engineering courses, which suffered a major drop in popularity from 65% of departments requiring the courses in 1989 to only 48% in 1994; material science maintained its position at about 46%.

The chemical engineering component constitutes 65% of the engineering category, a drop of 5% from the previous survey. Although transport phenomena and unit operations do overlap in course content to a considerable extent, they suggest a difference in focus. The number of departments

TABLE 2
Distribution of Course Work

Curricular Area	AIChE %	1981 Avg	1985 Avg	1989 Avg	1994 Avg
Mathematics beyond Trigonometry	12.5	13.6	12.7	12.4	12.5
Basic Sciences	25.0	24.3	25.4	24.8	24.1
(Incl. Advanced Chemistry)	(12.5)	(11.7)	(12.8)	(12.3)	(11.9)
Engineering Sciences/Design	37.5	37.3	37.2	39.7	39.7
Humanities/Social Sciences	12.5	16.1	15.1	13.5	14.6
Other	12.5	8.7	9.7	9.6	8.9
Total Percent	100.0	100.0	100.0	100.0	100.0
Total Credit Hours		133.4	131.4	132.8	132.8

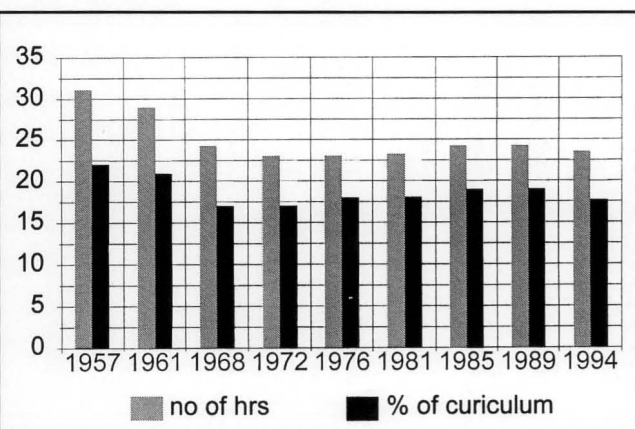


Figure 2. Chemistry content exclusive of advanced chemistry.

requiring unit operation theory decreased significantly from about 74% to 63%, but departments requiring unit operations laboratory is almost 94%.

In the case of transport phenomena, theory courses were reported by nearly 80% of the departments; the laboratory component was reported by only half of that number. Mass transfer is offered by 58%, and process control and process dynamics were reported by approximately 86% and 57% of the departments, respectively. Reactor design is required by three-quarters of the departments.

Regarding electives, twenty-one specific electives and a broad "other" choice were included in this category of the questionnaire. The results are given in Table 3. Biochemical, polymers, and the environmental electives continue to be the top three, with approximately 49% of the departments offering them. There has been a shuffling of positions between other electives. Equipment and energy related areas, such as natural gas and fuel, are still in the lower end.

The cultural content (which includes the humanities and the social sciences) has managed to break its declining trend of the past three surveys and is approximately equal to that of the 1985 survey (see Figure 3); it never actually reached the ABET minimum value of 12.5%. Interestingly, the range of credit hours required has narrowed, with the low end moving up from 6 to 13 hours and the high end moving significantly down from 55.3 to 37.36 hours. The high actually moved from 42% of the program to 28%, while the low end shifted from 5% to 10%.

The fifth and final major section of the questionnaire was classified as "other" and included diverse course offerings. The communication category formed a major portion of this section, but it has shown fluctuation over the years. In accordance with this fluctuating trend, it decreased from 90% to slightly less than 80%. As one would normally expect, computer programming (which is in this fifth, "other" category) was another course required by a significant number of the departments. Figure 4 shows results of responses for this category.

Table 4 depicts an average program that could be used for comparison. The information provided in this table might be useful to a school starting up a chemical engineering program.

Table 5 indicates that the average department reported 8% foreign undergraduate students, but 48% foreign graduate students. There are on the average 7.04 full professors, 2.59 associate professors, 1.83 assistant professors, and 0.62 full-time equivalent other faculty in the 63 schools that responded to the survey.

About 14% of salaries are obtained from other than general educational funds, and there are about 0.59 faculty positions available on the average. The number of faculty positions for the 63 reporting schools is about 37, but closer

TABLE 3
Elective Offerings

<i>Elective</i>	<i># Depts</i>	<i>%</i>	<i>Elective</i>	<i># Depts</i>	<i>%</i>
1. Biochemical	34	54.0	12. Petroleum	5	7.9
2. Polymers	29	46.0	13. Catalysts	6	9.5
3. Environmental	29	46.0	14. Paper	3	4.8
4. Transport Phenomena	11	17.5	15. Nuclear	4	6.3
5. Applied Mathematics	16	25.4	16. Coal	6	9.5
6. Control	13	20.6	17. Energy	4	6.3
7. Biomedical	13	20.6	18. Equipment	1	1.6
8. Design	8	12.7	19. Food	2	3.2
9. Mass Transfer	8	12.7	20. Fuel	2	3.2
10. Reactors	6	9.5	21. Natural Gas	2	3.2
11. Electrochemistry	10	15.9	22. Others	27	42.9

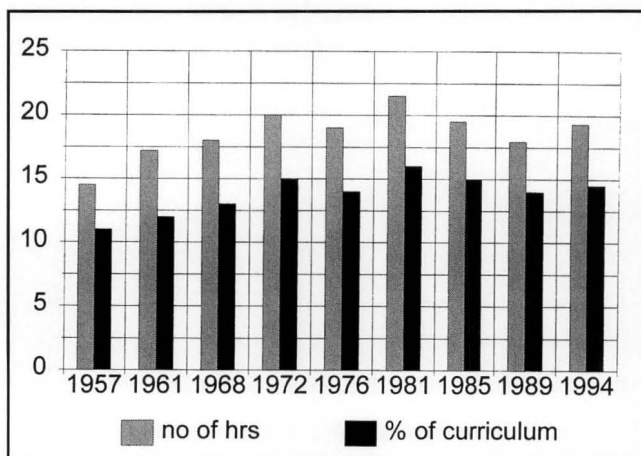


Figure 3. Cultural content.

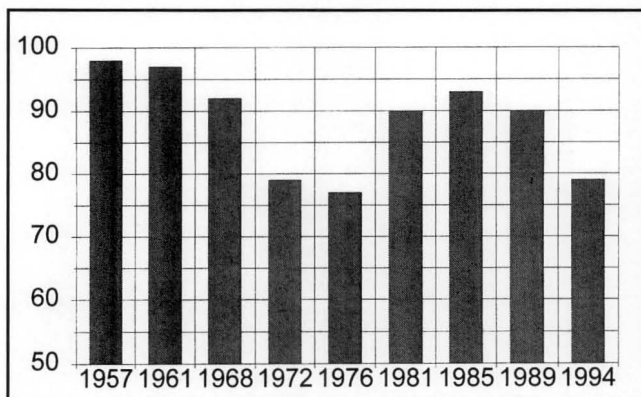


Figure 4. Communications (% of schools offering).

TABLE 4
Average Program Abstract

<u>Course</u>	<u>Hours</u>
Analytical Geometry	2.94
Calculus	8.31
Differential Equations	3.21
General Physics	7.67
General Chemistry	7.69
Physical Chemistry	6.36
Organic Chemistry	7.10
Other Chemistry	3.98
Statics	3.06
Electrical Engineering	3.77
Material Science	3.94
Fluid Mechanics	2.73
Heat Transfer	2.56
Material and Energy Balances	3.61
Thermodynamics	4.22
Reaction Engineering	1.88
Transport Phenomena	3.88
Mass Transfer	2.97
Unit Operations	3.20
Laboratory	3.74
Process Control	2.36
Design	4.93
ChE Electives	5.93
Humanities	9.72
Social Science	6.99
Communications	6.82
Computer Programming	2.58
Other	6.62
Total	132.77

TABLE 5
Summary of Student and Faculty Information

	<u>Total</u>	<u>Avg.</u>
<u>Students (ChE)</u>		
Fraction, Non-U.S. Undergraduate	N/A	0.08
Fraction, Non U.S. Graduate	N/A	0.48
<u>Faculty (Number of)</u>		
Full-time professors	443.63	7.04
Full-time associate professors	163.33	2.59
Full-time assistant professors	115.00	1.83
Full time equivalent, other teaching staff	39.3	0.62
% of salaries funded from other than general education funds	N/A	13.99
Number of full-time faculty positions open (tenure track)	37.00	0.59

understanding of the variation among departments necessitates a review of the entire information contained in the data received in all of the 63 responses.

The spreadsheets have been made available for all participating departments. For others who are interested, the spreadsheet will be made available upon request made in writing to the authors.

CONCLUSIONS

- Sixty-three departments of chemical engineering (out of 158 schools that were solicited) completed the most recent survey conducted by the Undergraduate Curriculum Subcommittee of the AIChE Education Projects Committee. The number of credits requirement for a BS degree in chemical engineering ranges from 115-145 on the ABET semester basis.
- The average number of credits required for a BS degree in the U.S. has remained almost the same at about 133 credits for the past twenty-two years.
- The chemistry content remained approximately the same for the past ten years, while the cultural content appears to have fluctuated the most for the past twenty years. The number of schools offering communications has seemed to decrease since 1985, although it had initially increased since 1976.
- The results in Table 2 indicate that the distribution of course work has remained fairly constant for the past thirteen years. There has been a slight increase in math courses beyond trigonometry, although it had decreased slightly and fairly steadily until 1989. Basic sciences has decreased slightly and fairly steadily for the past nine years; engineering science has increased in a similar fashion.
- The results in Table 3 indicate that biochemical electives are offered at the highest percentage of the schools.
- Table 4 provides an average program abstract of the course offerings and indicates little change since the last survey, performed in 1989.^[1]

ACKNOWLEDGMENTS

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REFERENCES

1. Coulman, G.A., "The Chemical Engineering Curriculum," *Chem. Eng. Ed.*, **23**(4), 184 (1990). Note; this reference contains a bibliography (seven related references) for this paper's topic. □

Random Thoughts . . .

IF YOU'VE GOT IT, FLAUNT IT *Uses and Abuses of Teaching Portfolios*

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A memo from the Provost appears in all faculty mail boxes one morning, announcing that from now on every candidate for tenure and promotion must submit a teaching portfolio along with the usual research documentation. Faculty reaction is swift and divided, even though no one understands exactly what is being required or why. Some professors see the requirement as an indication that the administration is finally starting to take teaching seriously, others view it as just another drain on their time that won't accomplish anything useful and could hurt them. Either viewpoint could turn out to be correct, depending on how the portfolio program is handled.

A **teaching portfolio** is a collection of materials that document a professor's teaching goals, strengths, and accomplishments. It contains

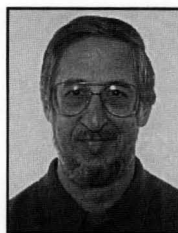
- ☐ *Self-generated material* (e.g., a teaching philosophy statement; representative syllabi, instructional objectives, handouts, assignments, and tests; descriptions of educational innovations and evaluations of their effectiveness; textbooks and education-related papers published; instructional software developed; teaching workshops and seminars presented or attended).
- ☐ *Teaching products* (e.g., graded assignments, tests, and reports; scores on standardized tests; student publications or presentations on course-related work).
- ☐ *Information generated by others* (e.g., summaries of student, alumni, and peer evaluations; honor and awards; reference letters).

Some items may be mandated, others may be included at the professor's option.

Portfolios have been used to document college teaching performance beginning in Canada in the 1970s, and their use

has become increasingly widespread since the 1991 publication of *The Teaching Portfolio* by Peter Seldin.^[1] Despite abundant evidence that their use improves teaching,^[2] the required inclusion of teaching portfolios in promotion and tenure dossiers is often viewed with faculty skepticism. This attitude may prove to be justified, as poorly designed or implemented portfolio programs are likely to have a minimal impact on institutional teaching quality and a negative impact on faculty morale. In the remainder of this column, we extract ideas from Seldin^[1,2] on ways to avoid the pitfalls and make portfolio programs effective.

What is the purpose of the teaching portfolio? • A portfolio can be used for *summative evaluation* (to evaluate teaching performance and provide a rational basis for promotion and tenure decisions and teaching award selections) or *formative evaluation* (to help identify and correct teaching problems). What goes in the portfolio depends on which function is intended. For summative evaluation, the portfolio should include some mandated items like a teaching



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philosophy and a summary of student ratings and some optional items that reflect on teaching performance and educational scholarship (e.g., student products, descriptions of teaching innovations, and reference letters from alumni or colleagues). For formative evaluation, the choice of content is entirely up to the professor and the focus should be on problem areas. The same portfolio should not be used for both functions.

How should summative portfolios be designed and evaluated?

• Three key requirements for effective portfolios are *relevance* (the elements selected for evaluation must be clearly linked to established criteria for effective teaching), *reliability* (ratings from different evaluators should be reasonably similar), and *practicality* (portfolios should be well organized, not too long, and easy to evaluate). While the ideal portfolio structure may vary considerably from one institution to another and from one discipline to another, the following design procedure is broadly applicable:

1. *Select categories that will be used to define the quality of a professor's teaching performance* (e.g., course design, instructional delivery, content expertise, development of new instructional methods and materials), *and assign relative weights to each category.*
2. *Formulate an objective set of questions addressing each category* (e.g., questions for the course design category might include, "Are the instructional objectives appropriate and consistent with the candidate's teaching philosophy and with institutional or departmental goals?" "Are the assignments and tests consistent with the objectives?")
2. *Specify required portfolio materials that will help provide meaningful answers to the questions.*

Once a summative portfolio has been prepared, several people should independently examine it, rate each category using a predefined system (e.g., 0=poor, 5=outstanding), calculate a weighted average rating, attempt to reconcile widely divergent evaluations, and finally provide a collective rating.

What is the point of the teaching philosophy statement?

• The philosophy statement enables portfolio evaluators to judge how well institutional goals and generally accepted criteria for good teaching are reflected in the professor's objectives, and the remaining portfolio contents can then be used to assess how well the objectives are being met. Good teaching is clearly being done when appropriate goals have been chosen and the portfolio contents demonstrate success in achieving them. Moreover, simply reflecting on *why* we do *what* we do in the classroom is likely to improve our teaching, even if the portfolio preparation goes no further.

How should new professors be assisted with portfolio preparation?

• There should be no secret about what

constitutes an outstanding portfolio and what constitutes an acceptable one. Discipline-specific model portfolios, like the illustrative ones given by Seldin,^[2] should be shown to professors at the outset of the process, and faculty colleagues or campus teaching consultants should be available as *portfolio mentors* to offer guidance and support. The mentors do not have to be in the same disciplines as the professors they are helping, but they should clearly understand the evaluation criteria used in those disciplines.

How should a portfolio program be initiated and institutionalized?

• Seldin cautions, repeatedly and emphatically, that *a portfolio program developed by administrators and imposed on the faculty will probably not achieve its objectives*, and suggests several ways to promote institutional acceptance.^[2] Administrators at all levels (department, school, and institution), in collaboration with the faculty, should set clear standards for both outstanding teaching and acceptable teaching, and they should publicize the portfolio evaluation criteria so that faculty members are clear about institutional expectations. The program should be pilot-tested on volunteers, including some of the most prestigious teachers and researchers on the faculty, before an attempt is made to institutionalize it. The administration should support portfolio development workshops and mentorships, e.g., by providing release time or other compensation for the workshop leaders and mentors.

Perhaps most importantly, the administration should demonstrate by actions as well as words its commitment to take portfolios seriously when making personnel decisions. If professors with strong teaching portfolios are treated the same as professors with strong research records in promotion and tenure decisions, faculty acceptance is likely to follow and the portfolio program has a good chance of working. Conversely, if professors with strong teaching portfolios and weak research records are denied tenure while others with weak teaching portfolios and strong research records get it, faculty acceptance will almost certainly be unattainable and the portfolio program is likely to fail.

This synopsis hardly does justice to the wealth of models and tips Seldin offers for portfolio preparation and evaluation. Anyone thinking about implementing a portfolio program should study the references and, if possible, attend a Seldin workshop. The potential impact of the program on teaching quality justifies doing whatever it takes to get it right the first time.

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2. Seldin, Peter, *Successful use of Teaching Portfolios*, Anker Publishing Company, Inc., Bolton, MA (1993) □